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Single Line

6. How does client-side routing work in react router?

Client-side routing means navigating between different pages or views in a react application without requesting a completly new HTML page from the server every time.

It most commonly used in react router navigation.

* Home
* Jobs
* My application
* My resume
* logsn.

* what is controlled component?

A controlled component is a form element whose value is controlled by react state.

* useState() to store input values.
* onChange to capture user input.
* value to display the current state.

+ Uncontrolled?

Uncontrolled is a form element whose value is managed by the browser DOM instead of react state.

9. How does react render list and conditional content?

react uses JS expressions inside JSX to display list and conditionally show or hide UI elements.

* list rendering
* conditional rendering.

The JS map() method creates a new array by applying a function to every element in an existing array.

here, keys are the special attribute.

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Syntax for useEffect:-

```
useEffect (() => {

    return () => {

    };
}, [dependencies]);
```

* cleanup function?

A cleanup function is used to remove or stop something created by an effect.

It prevents unnecessary background operations, memory leaks and unwanted behaviours.

8. Control & uncontrolled components

mainly it focus two approaches manage form inputs.

* login forms
* registration forms
* contact forms
* Job application forms
* search boxes.

Controlled - react manages the input value

uncontrolled - the browser Dom.

2) What are State

It is the data managed inside a react component.

State can change when a user performs an action, such as clicking a button, typing in an input, or selecting an option.

4) What is one-way data flow?

one-way flow means data moves in only one direction: P → C component.

5) Immutability in react?

It means we should not directly modify existing state variables. we should create a new value when update data.

6. How Does useState() work in react?

It is a react hook used to create and manage state in functional components.

State is data that can change at.

It shows the updated value.

Syntax for useState.

```
const [state, setState] = useState(value);
```

* what are state snapshots?.

A state snapshot means each render gets its own fixed view of the state values for that particular render.

when react renders a component, the state variables inside that render.

* what is batching react.

batching means react groups multiple state updates together and process them in a single render whenever possible.

7). How DOES useEffect() work, and when should it be used?

use effect() is a react hook used to handle side effects in functional components.

* Fetching data from API.
* setting a timer using setInterval()
* updating the browser document title
* Adding event listeners.
* connect or disconnect from external services.

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* State changes:

State is a data managed inside a react component.

When state changes using state function, react reschedules a re-render.

* Props changes:-

Props are data passed from a parent component to a child component.

When a parent passes new Props to child the child normally re-renders.

* Why re-rendering does not always mean DOM changes.

5. What are Props and state in react, and how are they different?.

Props:- used to pass data from a parent component to a child component.

State: Used to store and manage data inside a component that can change over time.

1) what are Props?

Props stands for Properties.

Props are used to pass data from one component to another, usually from parent to child.

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2. What is JSX? → JavaScript XML

It is a Syntax extension for JS that allows us to write HTML code inside JS.

It makes React code easier to read & understand.

* How does browser executes JSX?

JSX must first be converted into JS using tool like Babel or a JSX compiler.

* What is JSX Transformation?

It means converting JSX syntax into regular JS code that the JS engine can execute.

* What are JS expressions inside JSX?

A JS expression is a piece of code that produce a value. We use curly braces {}.

* How is JSX different from HTML?

| HTML | JSX |
|---|---|
| used to structure webpag | used to describe UI in React. |
| Use class | use className. |
| Use for | use htmlFor |
| Can Contain HTML standard | Must follow Syntax rules |

3. Real DOM

Real DOM is actual structure of HTML elements displayed in the browser.

* represents the actual HTML elements.

Virtual DOM.

Virtual DOM is a lightweight JS representation of the real DOM.

* react uses it to track UI changes and determine what needs to be updated in browser

* what is rendering in react.

rendering means creating the UI based on the current Props and state of a component.

* what is reconciliation?

reconciliation is the process React uses to compare the previous UI representation with the new UI representation and determine what changes are needed.

4. what cause a react components to render or re-render?

* React: react creates the UI based on a component current data (State and PROPS).

* Re-render: react executes the components function to calculate the updated UI.

In react, a component can render or re-render for four main reasons.

* What is Component - Based architecture?

Component - based architecture means dividing a large website into small, independent and reusable. each component handle a particular part of the UI.

Ex:- Job Portal website.

* Navbar component
* Sidebar component.
* Search bar component
* Jobcard component
* Footer component.

* Declarative UI.

Declarative UI means telling react what the UI should look like, instead of manually explaining every step to update the webpage.

* What Problem does react solve?

Before react, developers often used JS Dom manipulation to update webpage elements manually.

10/11/26

# Task

M. Dhanaseelan.

1. What is react ?

react is a JS library used to build fast, interactive and reusable UI ) especially for web application.

* Developed and maintained by meta
* Used to build a Single Page applications.
* uses components to divide a webpage into smaller parts
* Uses JSX to write HTML-like code inside JS.

* why react called a library instead of a framework?

react is called library because it mainly focuses on building the user interface (UI)

react library

* mainly focuses on UI
* more flexibility
* Developers choose additional tools.

Framework:

It Provides a complete application structure.

* react handles the UI
* react router can handle navigations.
* Axsous
* Redux.